

NovaTech Solutions Pvt. Ltd.

NovaDesk REST API Documentation

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1. Introduction

The NovaDesk REST API enables authorized applications to securely integrate with the NovaDesk Enterprise Service Management Platform.

The API allows external systems to:

- Create service requests
- Retrieve tickets
- Update incidents
- Search knowledge articles
- Manage assets
- Retrieve reports
- Synchronize users
- Automate workflows

The API follows REST architectural principles and exchanges data using JSON over HTTPS. Only authenticated and authorized clients may access protected endpoints.

2. API Overview

The NovaDesk API is designed for:

- Internal integrations
- Third-party applications
- Enterprise middleware
- Mobile applications
- Reporting platforms
- Automation tools

Key characteristics include:

- RESTful architecture
- HTTPS only

- JSON request and response bodies
- OAuth 2.0 authentication
- API Key support for approved service accounts
- Standard HTTP status codes
- Predictable resource naming
- Versioned endpoints

The API is stateless. Every request must contain all required authentication and context information.

Supported HTTP Methods

Method Purpose

GET	Retrieve data
POST	Create resources
PUT	Replace existing resources
PATCH	Partially update resources
DELETE	Remove resources

Content Types

The API supports the following media types.

Request

Content-Type: application/json

Response

application/json

Other content types are not supported unless explicitly documented.

3. Base URL

Production Environment

<https://api.novatechsolutions.com/v3>

Sandbox Environment

<https://sandbox-api.novatechsolutions.com/v3>

The sandbox environment should be used for development and testing.
Production credentials must never be used within the sandbox environment.

HTTPS Requirement

All API requests must use HTTPS.
HTTP requests are automatically rejected.
TLS version 1.2 or later is required.

4. Authentication

NovaDesk supports two authentication mechanisms.

- OAuth 2.0
- API Keys (Service Accounts)

OAuth 2.0 is the recommended authentication method.

OAuth 2.0 Flow

The authentication sequence is:

1. Application requests authorization.
 2. User authenticates.
 3. Authorization code is returned.
 4. Application exchanges the code for an access token.
 5. Access token is included in subsequent API requests.
-

Access Token

Access tokens remain valid for:

60 minutes

After expiration, applications must obtain a new token using a refresh token or reauthenticate.

Refresh Token

Refresh tokens remain valid for:

30 days

Expired refresh tokens require the user to complete the authentication process again.

API Keys

Service accounts may authenticate using API keys.

API keys are intended for:

- Scheduled jobs
- System integrations
- Monitoring tools
- Enterprise middleware

API keys must never be embedded directly into client-side applications.

Authentication Header

Every authenticated request must include:

Authorization: Bearer <access_token>

Example

GET /tickets

Authorization: Bearer eyJhbGciOi...

Requests without valid authentication receive an HTTP **401 Unauthorized** response.

5. Authorization

Authentication verifies identity.

Authorization determines what the authenticated user may access.

NovaDesk uses Role-Based Access Control (RBAC).

Standard Roles

Role	Permissions
End User	View and create personal tickets
Support Agent	Manage assigned tickets

Role	Permissions
Team Lead	Manage team tickets
Administrator	Full platform administration
API Service Account	Configurable integration permissions

Permission Examples

Support Agents can:

- View assigned tickets
- Update ticket status
- Add internal notes
- Upload attachments

Support Agents cannot:

- Delete users
 - Modify system configuration
 - Access administrator-only endpoints
-

Forbidden Requests

Users attempting to access unauthorized resources receive:

403 Forbidden

6. HTTP Headers

Most API requests include the following headers.

Header	Required
Authorization	Yes
Content-Type	Yes
Accept	Yes
User-Agent	Recommended
X-Correlation-ID	Recommended

Example Request Headers

Authorization: Bearer <token>

Content-Type: application/json

Accept: application/json

User-Agent: NovaDesk-Client/3.2

X-Correlation-ID: 6f2f7f8a

Correlation ID

The optional **X-Correlation-ID** header enables request tracing across distributed systems.

If not supplied, NovaDesk automatically generates a unique correlation identifier.

7. API Versioning

NovaDesk uses URI-based versioning.

Example:

/v1

/v2

/v3

The current production version is:

v3

Major versions may introduce breaking changes.

Minor releases maintain backward compatibility.

Deprecated API versions remain supported for a minimum of **12 months** after a newer major version becomes generally available.

8. Rate Limiting

To protect platform stability, NovaDesk enforces API rate limits.

Standard Limits

Client Type	Limit
Authenticated User	1,000 requests/hour
Service Account	5,000 requests/hour
Enterprise Integration	20,000 requests/hour (contract dependent)

Rate Limit Headers

Responses include:

X-RateLimit-Limit

X-RateLimit-Remaining

X-RateLimit-Reset

Applications should monitor these headers to avoid exceeding rate limits.

Exceeded Rate Limit

Clients exceeding their limit receive:

429 Too Many Requests

The response includes the recommended retry time.

9. Request Format

All request bodies must use JSON.

Example

```
{  
  "title": "Unable to access VPN",  
  "priority": "P2",  
  "category": "Network",  
  "description": "VPN connection fails after authentication."  
}
```



```
}
```

Field names are case-sensitive.

Unknown properties are ignored unless otherwise specified.

Date Format

Dates use ISO 8601.

Example

2026-01-15T10:30:45Z

Time values are represented in Coordinated Universal Time (UTC).

10. Response Format

Successful responses return JSON.

Example

```
{
  "success": true,
  "data": {
    "ticketId": "INC-2026-001284",
    "status": "New"
  }
}
```

Error responses include an error object.

Example

```
{
  "success": false,
  "error": {
    "code": "INVALID_REQUEST",
    "message": "Priority value is invalid."
  }
}
```

11. Common HTTP Status Codes

Status Code Meaning

200	Request successful
201	Resource created
204	No content
400	Bad request
401	Authentication required
403	Access denied
404	Resource not found
409	Conflict
422	Validation error
429	Rate limit exceeded
500	Internal server error
503	Service unavailable

12. Pagination

Large result sets use pagination.

Default page size:

25 records

Maximum page size:

100 records

Example

GET /tickets?page=2&pageSize=25

Response metadata includes:

- Current Page
 - Page Size
 - Total Records
 - Total Pages
-

13. Filtering

Most collection endpoints support filtering.

Examples

GET /tickets?status=Open

GET /tickets?priority=P1

GET /tickets?category=Network

Multiple filters may be combined.

14. Sorting

Sorting is available on most list endpoints.

Ascending

sort=createdDate

Descending

sort=-createdDate

Supported sort fields include:

- createdDate
- updatedDate
- priority
- status
- assignee
- category

Only documented fields may be used for sorting.

15. Tickets API

The Tickets API allows applications to create, retrieve, update, search, and close service desk tickets.

Base Resource

/tickets

Create Ticket

Creates a new incident or service request.

Endpoint

POST /tickets

Required Permissions

- Support Agent
 - Administrator
 - API Service Account
-

Request Body

```
{  
  "title": "Unable to access VPN",  
  "description": "VPN authentication fails after entering credentials.",  
  "category": "Network",  
  "priority": "P2",  
  "requesterId": "USR-10231"  
}
```

Successful Response

Status

201 Created

```
{  
  "success": true,  
  "data": {  
    "ticketId": "INC-2026-004812",  
    "status": "New",  
    "createdDate": "2026-01-18T09:15:42Z"  
  }  
}
```

Retrieve Ticket

GET /tickets/{ticketId}

Example

GET /tickets/INC-2026-004812

Response

```
{
  "ticketId": "INC-2026-004812",
  "status": "In Progress",
  "priority": "P2",
  "category": "Network",
  "assignee": "John Smith",
  "createdDate": "2026-01-18T09:15:42Z",
  "updatedAt": "2026-01-18T10:05:14Z"
}
```

Update Ticket

PATCH /tickets/{ticketId}

Example

```
{
  "status": "Resolved"
}
```

Only supplied fields are updated.

Delete Ticket

Ticket deletion is restricted.

DELETE /tickets/{ticketId}

Only Administrators may permanently delete tickets.

Deleted tickets cannot be recovered.

Search Tickets

GET /tickets

Supported filters

status

priority

category

assignee

requester

createdAfter

createdBefore

Example

GET /tickets?status=Open&priority=P1

16. Users API

The Users API manages employee accounts.

Base Resource

/users

Retrieve User

GET /users/{userId}

Example

GET /users/USR-10231

Response

```
{  
  "userId": "USR-10231",
```

```
"displayName": "Rahul Sharma",  
"department": "Engineering",  
"role": "Support Agent",  
"email": "rahul.sharma@novatechsolutions.com",  
"status": "Active"  
}
```

Create User

POST /users

Required fields

- Display Name
- Email
- Department
- Role

Passwords are not supplied directly.

A temporary password is automatically generated.

Disable User

PATCH /users/{userId}

```
{  
  "status": "Disabled"  
}
```

Disabled users cannot authenticate.

17. Assets API

The Assets API manages hardware and software inventory.

Base Resource

/assets

Create Asset

POST /assets

Request

```
{  
  "assetName": "Dell Latitude 7450",  
  "category": "Laptop",  
  "serialNumber": "DL7450-98312",  
  "assignedTo": "USR-10231"  
}
```

Retrieve Asset

GET /assets/{assetId}

Response

```
{  
  "assetId": "AST-003291",  
  "status": "Assigned",  
  "assignedTo": "USR-10231",  
  "purchaseDate": "2025-04-10"  
}
```

Asset Status Values

Supported values

- Available
 - Assigned
 - Reserved
 - Maintenance
 - Retired
 - Lost
-

18. Knowledge Base API

Base Resource

/articles

Knowledge articles are searchable and version-controlled.

Retrieve Article

GET /articles/{articleId}

Response

```
{  
  "articleId":"KB-1024",  
  "title":"VPN Connection Troubleshooting",  
  "version":"4",  
  "category":"Networking"  
}
```

Search Articles

GET /articles/search

Example

GET /articles/search?q=vpn

Search supports

- Keywords
 - Tags
 - Categories
 - Author
 - Article ID
-

Publish Article

POST /articles

Requires

Administrator

or

Knowledge Manager permissions.

19. Reports API

Base Resource

/reports

Generate Report

POST /reports

Request

```
{  
  "report": "sla-compliance",  
  "format": "pdf"  
}
```

Supported formats

- pdf
 - csv
 - xlsx
-

Download Report

GET /reports/{reportId}

Downloaded reports remain available for

30 days

before automatic deletion.

20. Search API

The Global Search endpoint allows applications to search multiple resource types simultaneously.

Base Resource

/search

Example

GET /search?q=printer

Searches

- Tickets
 - Assets
 - Users
 - Knowledge Articles
-

Response

```
{  
  "tickets":14,  
  "articles":6,  
  "assets":3  
}
```

21. Workflow API

Workflow endpoints allow administrators to automate business processes.

Base Resource

/workflows

Create Workflow

POST /workflows

Required permissions

Administrator

Workflow definition includes

- Trigger
 - Conditions
 - Actions
-

Supported triggers

- Ticket Created
 - Ticket Updated
 - SLA Warning
 - Asset Assigned
 - User Created
-

Supported actions

- Send Email
 - Create Task
 - Assign Ticket
 - Notify Manager
 - Escalate Ticket
 - Update Status
-

22. Attachments API

Attachments may be uploaded to tickets and knowledge articles.

Base Resource

/attachments

Upload Attachment

POST /attachments

Supported formats

- PDF
- DOCX
- XLSX
- TXT
- PNG
- JPG
- ZIP

Maximum file size

50 MB

Download Attachment

GET /attachments/{attachmentId}

Unauthorized users receive

403 Forbidden

23. Pagination Example

Collection endpoints return

```
{  
  "page":1,  
  "pageSize":25,  
  "totalPages":42,  
  "totalRecords":1038,  
  "data":[]  
}
```

24. Filtering Example

Example

GET /tickets

?priority=P1

&status=Open

&category=Network

Multiple filters may be combined.

25. Sorting Example

Ascending

sort=createdDate

Descending

sort=-createdDate

Supported fields

- createdDate
 - updatedDate
 - priority
 - status
 - assignee
-

26. Best Practices

NovaTech recommends the following practices when integrating with the REST API:

- Always use HTTPS.
- Store access tokens securely.
- Refresh tokens before expiration.
- Handle HTTP error responses appropriately.
- Implement retry logic with exponential backoff for transient failures.
- Respect published rate limits.
- Avoid polling when webhook notifications are available.
- Validate all input data before sending requests.
- Use pagination for large datasets.
- Log request correlation IDs to simplify troubleshooting.

27. Error Handling

The NovaDesk REST API uses standard HTTP status codes together with structured JSON error responses.

Every failed request returns an error object describing the problem.

Example

```
{  
  "success": false,  
  "error": {
```

```

    "code": "INVALID_REQUEST",
    "message": "Priority value is invalid.",
    "correlationId": "7af3b6d1-44c2-4df6-9c52-1b72dfba2a45"
  }
}

```

The correlationId should be included when contacting the IT Service Desk or Platform Engineering team.

Standard Error Codes

Error Code	Description
INVALID_REQUEST	Request validation failed
INVALID_TOKEN	Authentication token is invalid
TOKEN_EXPIRED	Access token has expired
ACCESS_DENIED	User lacks required permissions
RESOURCE_NOT_FOUND	Requested resource does not exist
DUPLICATE_RESOURCE	Resource already exists
RATE_LIMIT_EXCEEDED	API rate limit exceeded
INTERNAL_ERROR	Unexpected server error
SERVICE_UNAVAILABLE	Service temporarily unavailable

Applications should avoid exposing internal error details directly to end users.

Retry Recommendations

Status Code	Retry Recommended
400	No
401	After obtaining a valid token
403	No
404	No
409	After resolving the conflict

429	Yes
500	Yes
503	Yes

For HTTP 429, clients should respect the Retry-After response header.

28. Webhooks

NovaDesk supports outbound webhooks to notify external systems when specific events occur. Webhooks reduce the need for continuous polling.

Supported Events

Event	Description
ticket.created	New ticket created
ticket.updated	Ticket updated
ticket.resolved	Ticket resolved
ticket.closed	Ticket closed
asset.assigned	Asset assigned
article.published	Knowledge article published
workflow.executed	Workflow completed
user.created	User account created

Webhook Payload

Example

```
{
  "event": "ticket.created",
  "timestamp": "2026-01-18T09:15:42Z",
  "data": {
    "ticketId": "INC-2026-004812",
    "priority": "P2",
    "status": "New"
  }
}
```



```
}  
}
```

Webhook payloads are delivered in JSON format over HTTPS.

Webhook Security

NovaDesk signs webhook requests using an HMAC SHA-256 signature.

Clients should verify the signature before processing the payload.

Webhook endpoints should always use HTTPS.

Retry Policy

If delivery fails:

- First retry: 1 minute
- Second retry: 5 minutes
- Third retry: 15 minutes
- Fourth retry: 1 hour
- Fifth retry: 24 hours

After the fifth unsuccessful attempt, the webhook is marked as failed and administrators are notified.

29. SDK Examples

NovaDesk provides sample code to simplify API integration.

cURL Example

Retrieve a ticket.

```
curl -X GET \  
https://api.novatechsolutions.com/v3/tickets/INC-2026-004812 \  
-H "Authorization: Bearer <access_token>"
```

Python Example

```
import requests
```

```
headers = {  
    "Authorization": "Bearer YOUR_ACCESS_TOKEN"  
}  
  
response = requests.get(  
    "https://api.novatechsolutions.com/v3/tickets/INC-2026-004812",  
    headers=headers  
)  
  
print(response.json())
```

JavaScript Example

```
fetch("https://api.novatechsolutions.com/v3/tickets/INC-2026-004812", {  
    headers: {  
        Authorization: "Bearer YOUR_ACCESS_TOKEN"  
    }  
})  
    .then(response => response.json())  
    .then(data => console.log(data));
```

30. Security Recommendations

Applications integrating with NovaDesk should follow secure development practices.

Recommended controls include:

- Store API credentials securely.
- Rotate API keys regularly.
- Use OAuth 2.0 whenever possible.
- Never hardcode secrets into source code.
- Encrypt sensitive data in transit.
- Validate all input.
- Verify webhook signatures.

- Log request correlation IDs.
- Apply the principle of least privilege.

API credentials should be stored in secure secret management systems rather than configuration files.

31. Monitoring and Logging

Applications should monitor API usage continuously.

Recommended metrics include:

- Requests per minute
- Average response time
- Error rate
- Authentication failures
- Rate-limit violations
- Webhook delivery success
- API latency

Logs should include:

- Timestamp
- Endpoint
- HTTP method
- Status code
- Correlation ID
- Client IP
- User or Service Account ID

Sensitive information such as passwords and access tokens must never be written to logs.

32. API Deprecation Policy

NovaDesk follows semantic versioning for API releases.

Major Versions

May include breaking changes.

Minor Versions

Add new functionality while maintaining backward compatibility.

Patch Versions

Provide bug fixes and security updates.

Deprecated endpoints remain supported for a minimum of **12 months** after the release of a newer major version.

Customers receive advance notice through:

- Product Release Notes
 - Customer Portal
 - Administrator Notifications
 - Email Announcements
-

33. Service Availability

NovaTech targets the following service objectives.

Metric	Target
API Availability	99.9% Monthly
Planned Maintenance Notice	Minimum 7 Days
Critical Incident Response	30 Minutes
Standard Support Response	4 Business Hours

Planned maintenance is normally performed outside standard business hours.

Emergency maintenance may occur immediately if required to protect platform security or stability.

34. Frequently Asked Questions

Q1. Which authentication method is recommended?

OAuth 2.0 is the recommended authentication mechanism for all applications.

Q2. How long is an access token valid?

Access tokens remain valid for **60 minutes**.

Q3. What should I do if I receive HTTP 429?

Reduce request frequency and retry after the interval specified in the Retry-After header.

Q4. Does the API support webhooks?

Yes. NovaDesk supports HTTPS webhooks for several resource events, including ticket, asset, article, workflow, and user events.

Q5. Which data format does the API use?

All requests and responses use JSON encoded with UTF-8.

Q6. What TLS version is required?

Clients must use **TLS 1.2 or later**.

Q7. How long are deprecated API versions supported?

Deprecated major API versions remain supported for at least **12 months** after the release of a newer major version.

Q8. What is the maximum page size?

The maximum supported page size is **100 records**.

Q9. What happens if webhook delivery fails?

NovaDesk retries delivery five times using an exponential retry schedule before marking the webhook as failed.

Q10. Where should API credentials be stored?

API credentials should be stored in a secure secret management system and never embedded directly in source code or client-side applications.

35. Glossary

Term	Definition
API	Application Programming Interface
OAuth 2.0	Authorization framework for secure API access
RBAC	Role-Based Access Control

SLA	Service Level Agreement
HMAC	Hash-based Message Authentication Code
JSON	JavaScript Object Notation
REST	Representational State Transfer
HTTPS	Secure Hypertext Transfer Protocol
UTC	Coordinated Universal Time

Revision History

Version	Date	Description
1.0	January 2024	Initial API Documentation
1.5	July 2025	Added Asset and Workflow APIs
2.0	January 2026	Added Webhooks, SDK Examples, Security Guidance, and Operational Policies

Related Documents

- NovaDesk User Guide (PROD-ND-001)
- NovaDesk Administrator Guide (PROD-ND-ADM-002)
- NovaDesk Release Notes (PROD-ND-004)
- Information Security Policy (COMP-SEC-003)
- Employee Handbook (HR-EMP-001)